CSE303: Statistics for Data Science
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Project Report

Exploratory Data Analysis of Superstore Sales Dataset

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1. Dataset Description

The dataset used in this project is a retail store dataset containing transactional data, customer details, shipping information, product categories and sales values. We can analyze business performance, customer behavior and sales trends using these data. This dataset represents realworld retail transactions, making it suitable for business analytics and decision-making studies.

- The dataset contains 9800 observations(rows) and 18 features(columns). The dataset was obtained from Kaggle.
- A brief description for the features in our dataset is as following:

Feature	Data type	Meaning	Category
Row ID	int64	Unique identifier for each row	Numerical (Discrete)
Order ID	string	Unique order identifier	Categorical (Nominal)
Order Date	string	Date when order was placed	Temporal (Date/Time)
Ship Date	string	Date when order was shipped	Temporal (Date/Time)
Ship Mode	string	Shipping method	Categorical (Nominal)
Customer ID	string	Unique customer ID	Categorical (Nominal)
Customer Name	string	Customer name	Categorical (Nominal)
Segment	string	Customer segment	Categorical (Nominal)
Country	string	Country	Categorical (Nominal)
City	string	City	Categorical (Nominal)
State	string	State	Categorical (Nominal)

Postal Code	float64	Postal code	Categorical (Nominal)
Region	string	Region	Categorical (Nominal)
Product ID	string	Product identifier	Categorical (Nominal)
Category	string	Product category	Categorical (Nominal)
Sub-Category	string	Product sub-category	Categorical (Nominal)
Product Name	string	Name of product	Categorical (Nominal)
Sales	float64	Sales amount	Numerical (Continuous)

Most variables in the dataset are categorical, indicating that the dataset is rich in descriptive attributes. Only a few numerical variables are present, with Sales being the primary continuous variable for quantitative analysis. Additionally, categorical variables dominate the dataset, while numerical analysis is primarily focused on the Sales feature. Overall, the dataset appears suitable for further exploratory analysis and predictive modeling due to its completeness and variety of features.

The Postal Code feature contains missing values (9789 non-null out of 9800). Date features are currently stored as strings and may need conversion to datetime format for further analysis. Missing value analysis shows that the dataset is highly complete. Among all features, only the Postal Code variable contains missing values, accounting for approximately 0.11% of the total data. All other variables have no missing entries. Since the proportion of missing data is very small, it is unlikely to significantly affect the overall analysis.

2. Central Tendency and Dispersion Analysis

Computational data of mean, median, mode, variance, standard deviation, and IQR are given below

Feature	Sales
Mean	230.769
Median	54.49

Mode	12.96
Variance	392692.572
Standard deviation	626.652
IQR	193.357

Interpretation:

The Sales feature shows that the mean is higher than the median, indicating a right-skewed distribution. This suggests the presence of high-value sales acting as outliers. The large difference between mean and median further confirms the presence of extreme values. The relatively high standard deviation reflects considerable variability in sales values. Furthermore, the IQR indicates that the central 50% of the data spans a wide range, reinforcing the presence of dispersion in the dataset.

3. Distribution and Shape Analysis

In this dataset, the most meaningful numerical feature is Sales (continuous variable). Other numeric fields like Row ID are identifiers, so they are not useful for distribution analysis.

To understand the distribution of Sales, the following visualizations should be included:

Histogram :

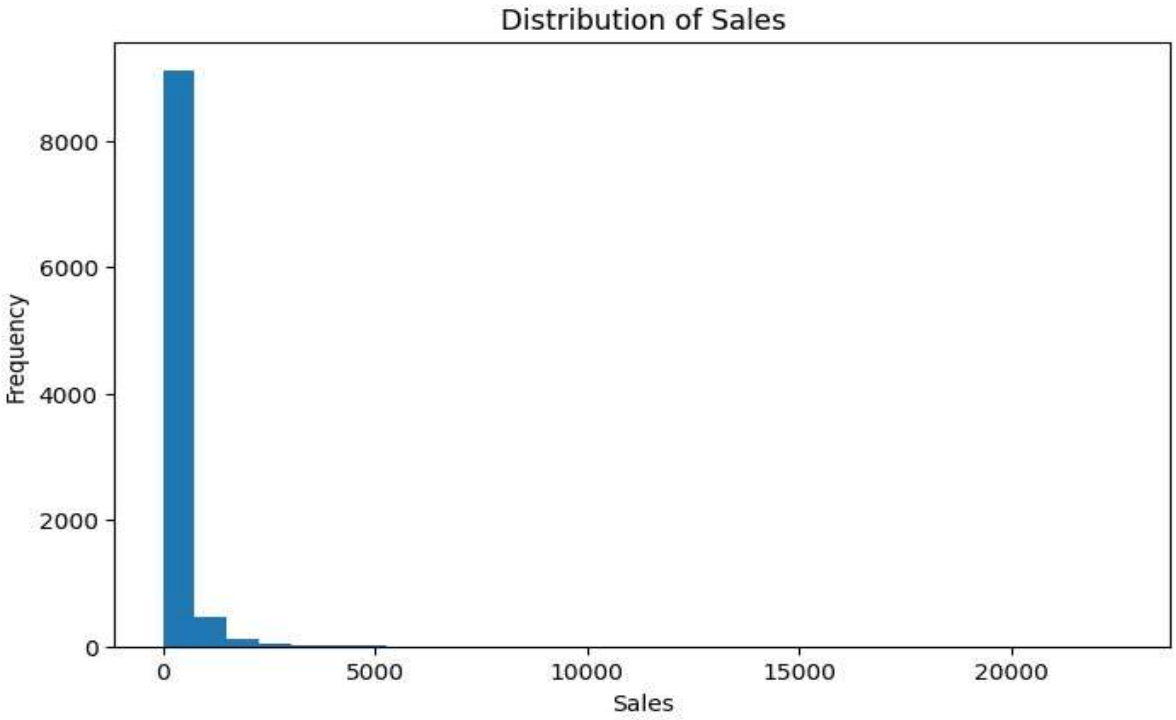


Figure 1: Histogram of Sales

Density Plot :

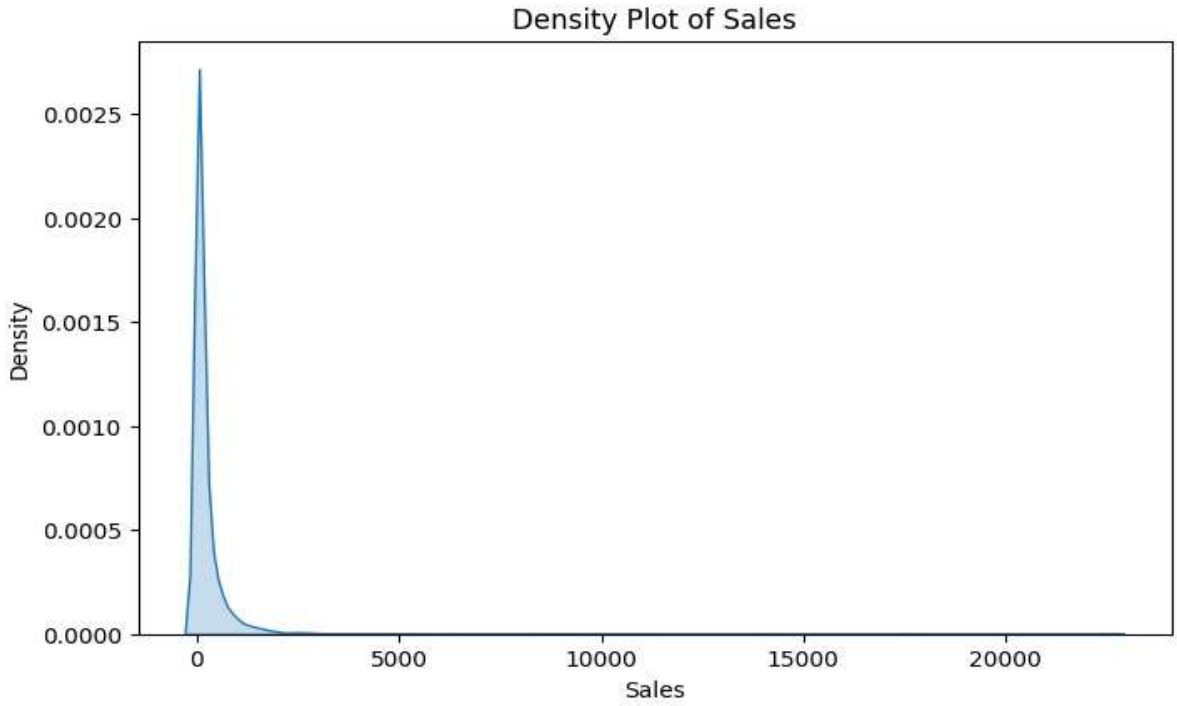


Figure 2: Density Plot

Boxplot :

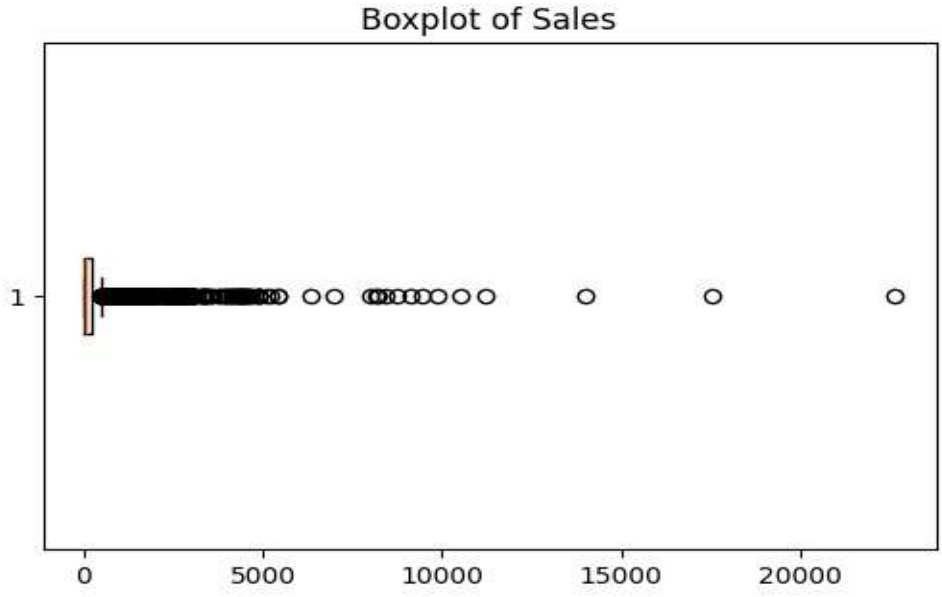


Figure 3: Boxplot of Sales

Shape of Distribution

The distribution of **Sales** is highly right-skewed, as indicated by the very high skewness value (≈ 12.98). This means that the majority of transactions are concentrated at lower sales values, while a small number of transactions have exceptionally high sales amounts, creating a long right tail.

Interpretation:

- Most orders in the dataset are low-value transactions**,** indicating typical customer purchasing behavior.
- A small proportion of orders contribute very high sales values, which significantly stretch the distribution.
- Due to this skewness, the mean is greater than the median, as it is influenced by extreme high values.
- This pattern suggests the presence of high-value customers or bulk purchases, which are not frequent but have a strong impact on overall sales.

Outlier Detection

To detect outliers, the Interquartile Range (IQR) method is used.

Metric	Value
Q1 (25th percentile)	17.25
Q3 (75th percentile)	210.61
IQR	193.36
Lower Bound	-272.79
Upper Bound	500.64

Interpretation:

- The large number of outliers indicates that extreme high sales values are not rare, but rather a consistent characteristic of the dataset.
- These outliers confirm that the distribution is heavy-tailed, meaning a significant portion of total sales comes from a small number of high-value transactions.

- Such extreme values can strongly influence statistical measures like the mean and standard deviation, potentially distorting analysis.
- For future modeling or analysis, it is important to handle these outliers carefully (e.g., transformation, capping, or separate analysis), depending on the objective.

4. Categorical Variable Analysis

Categorical Columns:

['Order ID', 'Order Date', 'Ship Date', 'Ship Mode', 'Customer ID', 'Customer Name', 'Segment', 'Country', 'City', 'State', 'Region', 'Product ID', 'Category', 'Sub-Category', 'Product Name']

Some key categorical variables include:

Segment ,Category ,Region ,Ship Mode

Frequency Distribution of Key Categorical Variables

Segment:

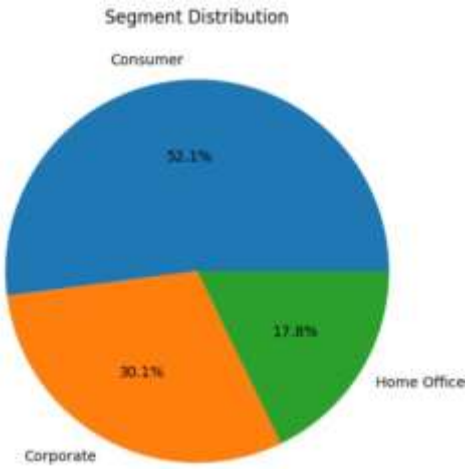
Segment	Count
Consumer	5101
Corporate	2953
Home office	1746

Table: Segment

- The Consumer segment has the highest number of transactions (5101), followed by Corporate and Home Office.
- This indicates that the business is primarily driven by individual customers rather than organizations.

Imbalance:

- There is a moderate imbalance, as Consumer dominates a large portion of the data compared to Home Office(≈52%).



Category:

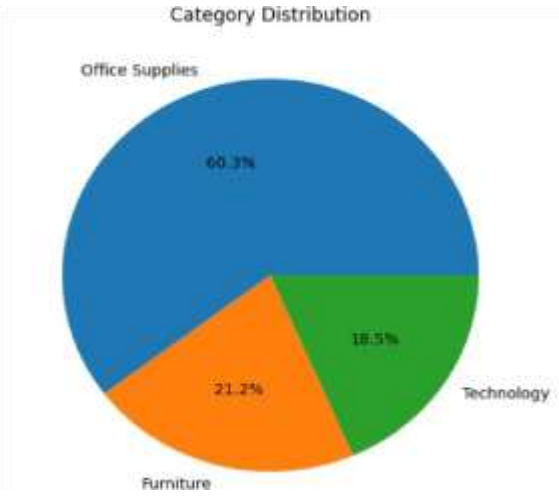
Category	Count
Office Supplies	5909
Furniture	2078
Technology	1813

Table: Category

- Office Supplies has a significantly higher count (5909) compared to Furniture and Technology.
- This suggests that customers frequently purchase low-cost, everyday items rather than expensive or specialized products.

Imbalance:

- There is a strong imbalance, with Office Supplies heavily dominating the dataset.



Region:

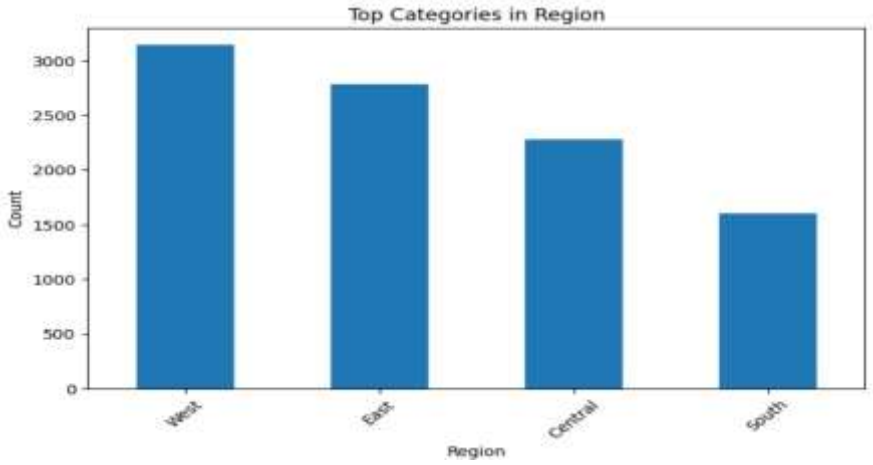
Region	Count
West	3140
East	2785
Central	2277
South	1598

Table: Region

- The West region has the highest number of orders, followed closely by East.
- South has the lowest number of transactions.

Imbalance:

- The distribution is relatively balanced, though West has a slight dominance.



Ship Mode:

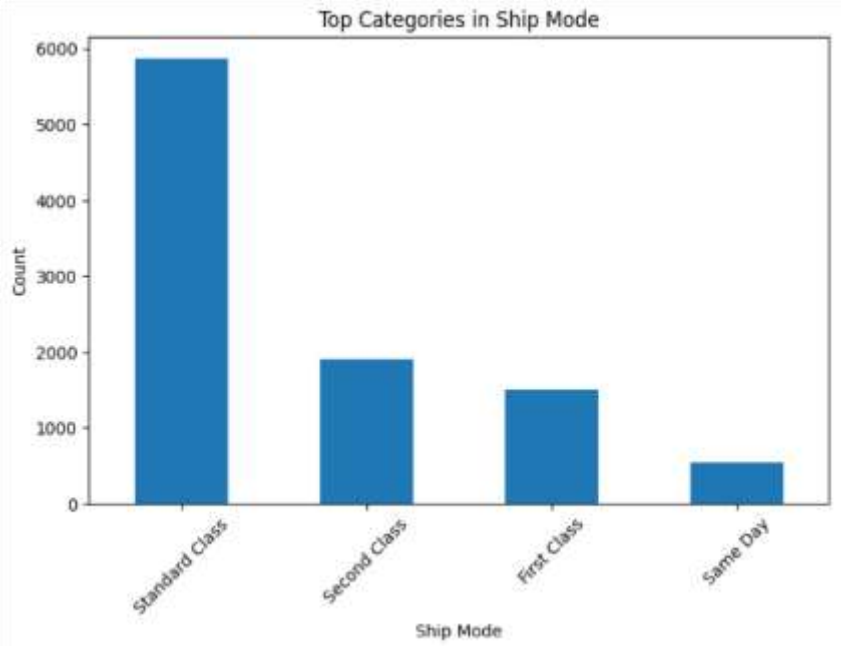
Ship Mode	Count
Standard Class	5859
Second Class	1902
First Class	1501
Same Day	538

Table: Ship Mode

- Standard Class is overwhelmingly the most used shipping mode (5859).
- Faster shipping options like Same Day are used much less frequently.

Imbalance:

- There is a clear imbalance, with Standard Class dominating(~60%), indicating customers prefer cost-effective shipping over speed.



5. Relationships Between Variables

In this dataset, Sales is the only meaningful numerical variable. To analyze its relationship with time, the Order Date is converted into a **Year-Month** format, creating a new column Year_Month where each date represents the corresponding month. This allows us to observe how sales vary over time.

A scatter plot is created with Year_Month on the X-axis and Sales on the Y-axis, where each point represents an individual transaction.

Observations:

- Sales show high variability across all months, with most transactions being low-value and a few extremely high-value orders (outliers).
- There is no consistent increasing or decreasing trend, but the scatter plot clearly shows monthly fluctuations in sales.

- Some months contain noticeable high-value transactions, highlighting the impact of outliers on overall sales.

Interpretation:

While the scatter plot does not indicate a strong linear relationship, it effectively visualizes how sales behave over time, helping identify patterns, fluctuations, and extreme transactions that may influence business analysis and modeling.



Figure: Scatter Plot

6. Data Quality Assessment

A thorough check of the dataset was performed to identify missing values, duplicate rows, and unusual values.

Columns	Missing Value percentage	Missing Value Count
Postal Code	0.112245	11
Order ID	0	0
Row ID	0	0
Ship Date	0	0
Ship Mode	0	0
Customer ID	0	0
Customer Name	0	0
Segment	0	0
Country	0	0
City	0	0

Order Date	0	0
State	0	0
Region	0	0
Product ID	0	0
Category	0	0
Sub-Category	0	0
Product Name	0	0
Sales	0	0
Year_Month	0	0

Missing Values:

- Only the Postal Code column has a small number of missing values (11 rows, 0.11% of the dataset).
- All other columns, including key numerical and categorical features, have no missing values.

Duplicate Rows:

- The dataset contains no duplicate rows, indicating that each record is unique and there is no redundancy.

Unusual Values (Shipping Days):

- Shipping days were calculated as the difference between Ship Date and Order Date.
- Summary statistics show shipping days range from 0 to 7 days, with a median of 4 days.
- No negative values were detected, confirming that shipping dates are consistent and valid.

Interpretation & Implications:

- Overall, the dataset has excellent quality, with only minimal missing values in Postal Code that are unlikely to impact most analyses.
- The absence of duplicates ensures accuracy in aggregations, statistics, and modeling.
- No unusual or inconsistent shipping values indicate reliable temporal data for analysis.
- The small number of missing Postal Codes can be handled by imputation (e.g., using city/state information) or leaving them blank if not critical.

7. Linear Regression

1. Data Splitting into Training and Testing Sets

In this step, the dataset is divided into two parts: a **training set** and a **testing set**. This is an essential step in machine learning to evaluate how well the model performs on unseen data.

Using the train_test_split function, 80%(7840) of the data is used for training the model, while the remaining 20%(1960) is reserved for testing. The random_state=42 parameter ensures that the split is reproducible. After splitting, the indices of both datasets are reset to maintain a clean and sequential order. Finally, both datasets are saved as separate CSV files for future use.

Dataset Type	Rows	Columns	Purpose
Training Set	7840	18	Model Training
Test Set	1960	18	Model Evaluation

2. Outlier Detection and Removal Using IQR Method

Outliers in the dataset are identified and removed using the Interquartile Range (IQR) method, a widely used statistical technique for detecting extreme values. Outliers can significantly distort the learning process of machine learning models, leading to inaccurate predictions.

The IQR method determines the first quartile (Q1) and third quartile (Q3), and calculates the interquartile range (IQR = Q3 − Q1). A lower and upper bound are then established using this range, and any observations falling outside these bounds are treated as outliers and removed from the dataset.The outlier removal process is primarily applied to the training dataset to enhance model performance and stability.

Dataset Type	Before Cleaning	After Cleaning	Rows remove
Training Set	7840	6956	884
Test Set	1960	1698	262

3. Feature Encoding and Data Preparation for Model Training

Categorical variables in the dataset are transformed into a numerical format using one-hot encoding, enabling their use in machine learning models. The selected categorical features include Category, Sub-Category, Region, Segment, and Ship Mode. To avoid redundancy and multicollinearity, the first category from each variable is dropped. The encoded features are then used to construct the feature matrix (X), while the target variable (y) is defined as the Sales column.

Subsequently, the dataset is divided into training and testing sets using an 80:20 ratio. This ensures that the model is trained on one portion of the data and evaluated on unseen data. To improve model performance and ensure numerical stability, feature scaling is applied using the StandardScaler.

Component	Shape	Description
X_train	(7840,26)	Training features
y_train	(7840,)	Training target (Sales)
X_test	(1960, 26)	Testing features
y_test	(1960,)	Testing target (Sales)

4. Bias Term Addition and Design Matrix Construction

A bias (intercept) term is introduced into the feature matrix to allow the linear regression model to estimate an intercept value. This is a fundamental requirement of the Normal Equation, as it enables the regression model to adjust vertically and achieve a better fit to the data.

To implement this, a column of ones is generated for both the training and testing datasets, representing the intercept term. This column is then horizontally concatenated with the original feature matrix to construct the final design matrices used for model training and evaluation.

The resulting matrices, denoted as A_train and A_test, include both the bias term and the feature variables, ensuring proper formulation for linear regression using the Normal Equation.

$$\text{Sales} = \beta_0 + \beta_1 \cdot x_1 + \beta_2 \cdot x_2 + \cdots + \beta_n \cdot x_n$$

Here,

β_0 = Intercept Term, β_1, β_2 = features's effect on output

Component	Shape	Description
X_train	(7840, 27)	Training design matrix with bias

y_train	(1960, 27)	Testing design matrix with bias
Bias Column	(7840,1) / (1960,1)	Intercept term (all ones)

The first row of the training design matrix (X_train) is shown below as a sample observation(Output):

[1. 1. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 1. 0. 0. 0. 0. 0. 0. 1. 0. 0. 0. 1. 0.]

This row represents a single training instance where:

- The first value (**1**) corresponds to the bias (intercept) term.
- The remaining values represent the encoded feature values obtained after one-hot encoding of categorical variables.
- Each binary value (0/1) indicates the presence or absence of a specific category feature.

5. Linear Regression Model Training using Normal Equation

The linear regression model is trained using the Normal Equation method, which provides an analytical solution for estimating the optimal model parameters. This approach eliminates the need for iterative optimization techniques such as gradient descent.

The parameter vector **β (beta)** is computed using the following matrix formulation:

$$B = (A^T A)^{-1} A^T y$$

Here, **A** represents the design matrix containing both the bias term and feature variables, while **y** represents the target variable (Sales). The resulting coefficients represent the relationship between each feature and the target variable, while the intercept captures the baseline prediction when all feature values are zero.

Intercept (β_0) : -17097.59

Feature **Coefficient**

Category -50975.31

Sub-Category 15578.51

Region 70143.66

Segment 72144.41

Ship Mode 68085.62

6. Model Training and Prediction

In this stage, a Linear Regression model from the Scikit-learn library is trained using the prepared feature matrix and target variable. The model learns the relationship between input features and the target variable (Sales) by minimizing the error between actual and predicted values.

After training, the model is used to generate predictions on the training dataset to evaluate its learning behavior. The predicted values are then compared with the actual values to assess model performance and consistency.

Observation	Actual Sales	Predicted Sales
1	49	60
2	128	521
3	240	233
4	469	198
5	56	49

7. Model Evaluation using Mean Squared Error (MSE) and Root Mean Squared Error (RMSE)

The trained linear regression model is evaluated on the test dataset to measure its predictive performance. Evaluation is performed using two standard regression metrics: Mean Squared Error (MSE) and Root Mean Squared Error (RMSE).
Mean Squared Error calculates the average of squared differences between actual and predicted values, providing a measure of overall prediction error
RMSE represents the average magnitude of prediction error in the same unit as the target variable (Sales), making it more interpretable for real-world analysis.

Test Mean Squared Error (MSE) : 4,645,837.16
Test Root Mean Squared Error (RMSE) : 2,155.42 (dollars)

An RMSE of 2155.42 indicates that, on average, the model's predictions deviate from actual sales by approximately $2155.

This suggests that while the model captures general trends, there is still room for improvement in predictive accuracy.

8. Actual vs Predicted Values Visualization

To visually evaluate the performance of the regression model, a scatter plot is used to compare the actual and predicted values on the test dataset. This graphical representation helps in understanding how closely the model predictions align with the true values.

Each point in the plot represents an observation, where the x-axis shows the actual values and the y-axis shows the predicted values. A reference diagonal line (y = x) is also included to indicate perfect prediction. The closer the points are to the diagonal line, the better the model performance.

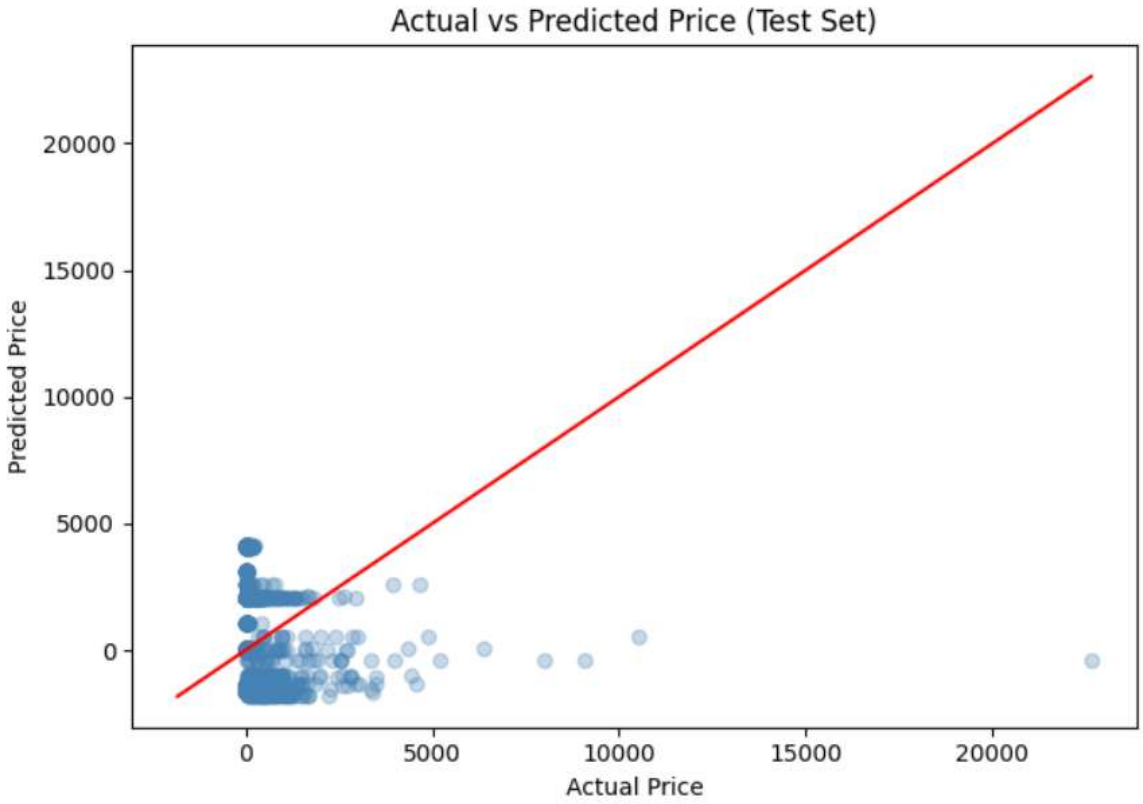


Figure: Actual vs Predicted Graph

9. Model Performance Comparison: With vs Without Categorical Features

The performance of the linear regression model is evaluated under two different scenarios: with categorical features and without categorical features. This comparison is conducted using Root Mean Squared Error (RMSE), which measures the average prediction error in the same unit as the target variable (Sales).

In the first case, only a baseline model is used without incorporating any categorical information. In the second case, categorical variables such as Category, Sub-Category, Region, Segment, and Ship Mode are included using one-hot encoding, allowing the model to capture more meaningful patterns from the data.

Evaluation Results

With Categorical Features — RMSE: 740.73 (dollars)

Without Categorical Features — RMSE: 818.99 (dollars)

8. Key Insights

1. **Sales Distribution and Outliers**

The Sales variable is right-skewed, as most values are relatively low (mainly below 200–500), while a few values are extremely high (above 20,000). This large difference between typical values and the maximum clearly indicates the presence of outliers, representing high-value transactions.

2. **Variability in Sales**

The standard deviation is high compared to the mean, showing that sales values are widely spread. This means transactions vary significantly, from small purchases to very large ones.

3. **Categorical Imbalance**

The dataset is imbalanced across categories. For example, Office Supplies appears much more frequently than Furniture and Technology. Similarly, the Consumer segment dominates over Corporate and Home Office, indicating most sales come from a specific group.

4. **Regional Differences**

Sales are not evenly distributed across regions. Some regions (e.g., West) have more transactions than others (e.g., South), showing geographical variation in sales performance.

5. **Temporal Patterns**

Sales show a slight increasing trend over the years, with more high-value transactions in later years. Some months also show higher activity, suggesting possible seasonal effects.

6. Shipping Performance

Most ShippingDays values fall between 2 to 5 days, with a maximum around 7 days. This indicates that deliveries are generally fast and consistent, with no major delays or errors.

7. Data Quality

The dataset is clean, with only a very small amount of missing data in Postal Code (around 0.11%) and no significant duplicate rows, ensuring reliable analysis.

8. Data Splitting

This step ensures that the model is trained and evaluated properly, preventing overfitting and providing a realistic estimate of model performance on new data.

9. Outlier Detection and Removal

Removing outliers helps the model focus on normal data patterns and reduces the impact of extreme values. This improves the stability and accuracy of the model.

10. Feature Encoding and Data Preparation

One-hot encoding increases the number of features (from categorical values to 26 numerical columns). Feature scaling ensures that all features contribute equally to the model, improving stability and performance.

11. Bias Term Addition

A bias term is added to the feature matrix to enable the model to learn an intercept, while the remaining encoded features represent how different categories influence sales predictions.

12. Linear Regression Model Training

The intercept (β_0) represents the baseline Sales value when all features are zero. Positive coefficients indicate a positive relationship with Sales, while negative coefficients indicate a negative impact. Features such as Region, Segment, and Ship Mode show strong positive influence on Sales, suggesting their importance in predicting revenue.

13. Model Training and Prediction

The model shows reasonable approximation for some observations, while larger deviations in certain cases indicate the presence of prediction error. This difference between actual and predicted values highlights the limitations of linear regression in capturing complex patterns. Despite some errors, the model successfully learns general trends in the dataset.

14. Model Evaluation

MSE represents the average squared deviation between actual and predicted values, heavily penalizing large errors. RMSE provides a more interpretable measure as it is expressed in the same unit as Sales (dollars).

15. Actual vs Predicted Values Visualization

Points clustered near the diagonal line indicate that the model has learned the underlying pattern effectively. Deviations from the line represent prediction errors. The spread of points suggests that while the model captures general trends, some variance still exists in predictions.

16. Model Performance Comparison

The model with categorical features performs significantly better, as indicated by the lower RMSE value. This confirms that categorical variables contain important information that improves predictive accuracy.

Important Variables for Future Analysis

The most important variable is **Sales**, as it serves as the target for prediction.

Category, Segment, and Region are useful because they influence sales patterns and show clear differences in distribution.

OrderMonth and OrderYear help capture trends and seasonality, while **ShippingDays** is useful for analyzing delivery performance and its potential impact on sales.